

Intestinal permeability and its relevance for absorption and elimination

H. LENNERNÄS

Biopharmaceutics, Uppsala University, Uppsala, Sweden

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Abstract

Human jejunal permeability (P_{eff}) is determined in the intestinal region with the highest expression of carrier proteins and largest surface area. Intestinal P_{eff} are often based on multiple parallel transport processes. Site-specific jejunal P_{eff} cannot reflect the permeability along the intestinal tract, but they are useful for approximating the fraction oral dose absorbed. It seems like drugs with a jejunal $P_{\text{eff}} > 1.5 \times 10^{-4} \text{ cm s}^{-1}$ will be completely absorbed no matter which transport mechanism(s) are utilized. Many drugs that are significantly effluxed *in vitro* have a rapid and complete intestinal absorption (i.e. >85%) mediated by passive transcellular diffusion. The determined jejunal P_{eff} for drugs transported mainly by absorptive carriers (such as peptide and amino acid transporters) will accurately predict the fraction of the dose absorbed as a consequence of the regional expression. The data also show that: (1) the human intestinal epithelium has a large resistance towards large and hydrophilic compounds; and (2) the paracellular route has a low contribution for compounds larger than approximately molecular weight 200. There is a need for more exploratory *in vivo* studies to clarify drug absorption and first-pass extraction along the intestine. One is encouraged to develop *in vivo* perfusion techniques for more distal parts of the gastrointestinal tract in humans. This would stimulate the development of more relevant and complex *in vitro* absorption models and form the basis for an accurate physiologically based pharmacokinetic modelling of oral drug absorption.

Keywords: Intestinal permeability, intestinal secretion, intestinal efflux, drug absorption, bioavailability, biopharmaceutics classification system, intestinal transporters, pharmacology, pharmacokinetics, intestinal perfusion, P-glycoprotein

Introduction

The gastrointestinal tract has several important functions besides the absorption and secretion of drugs. It must also absorb nutrients rapidly and at the same time act as an efficient barrier against potentially hazardous bacteria and toxins. Further, gut-associated

lymphoid tissue (GALT), which is the largest immunological tissue in the body, is located in the intestine (Brayden et al. 2005). This means that a single epithelial cell layer in the intestine, comprised mainly of enterocytes and colonocytes, performs the crucial task of regulating the balance between efficient absorption and optimal protection and, therefore, affects intestinal absorptive permeability and the secretion of drugs. The microflora in the intestinal lumen is essential for correct intestinal function and therefore plays an important role in health and disease. It has been estimated that the human body plays host to 10^{14} bacteria, ten times more than the 'resident' eukaryotic cells. About 90% of the cells in a body, therefore, are bacteria with the main part located in the lumen of the intestine (Luckey 1972). While some bacteria are found on the skin, in the respiratory tract and vagina, the vast majority is located within the gastrointestinal tract (which also is defined as outside the body). These bacteria form a complex ecosystem that influences the host and is a prerequisite for health and normal physiology. The microflora differ in composition and concentration along the gastrointestinal tract (Holzapfel et al. 1998). The stomach and the jejunum contain few bacteria due to low pH and inhibitory action by bile/pancreatic secretion, respectively. The composition of the microflora in the ileum approaches that of the colonic microflora. The concentration of bacteria in the colon is high due to a higher pH, an abundance of nutrients and less aggressive peristalsis. The human colon may be regarded as a continuously running fermentor generating 10^{12} organisms per gram of faeces. The fermentation by intestinal flora of non-degradable carbohydrates, such as non-starch polysaccharides and oligosaccharides, produces short-chain fatty acids; predominantly butyrate, propionate, and acetate. These fatty acids are absorbed by the epithelium and are metabolized in the colonic epithelium, liver, and peripheral tissue (Porter et al. 2007). The epithelium in the colon receives approximately 70% of its energy supply from fatty acids in the colonic lumen (Wong et al. 2006). The diverse physiological processes of the gastrointestinal tract are finely tuned and in balance to optimize the well-being of the host and play a highly significant role in the absorption and bioavailability of drugs.

An improved understanding of drug membrane transport in the small and large intestine in humans is vital to the development of new and effective oral pharmaceutical products (Ungell et al. 1998; Wilding et al. 2000; Davis and Wilding 2001; Brayden 2003; Lindahl et al. 2004; Lennernäs and Abrahamsson 2005; Petri et al. 2006a; Hinderling et al. 2007; Porter et al. 2007). New candidate drugs and old drugs in novel dosage forms often need to be examined better for their intestinal permeabilities, which often are based on multiple parallel transport processes such as passive transcellular diffusion, carrier-mediated absorption, and carrier-mediated efflux (Fagerholm et al. 1997; Ungell et al. 1998; Bourdet et al. 2005, 2006; Bourdet and Thakker 2006; Cao et al. 2006; Hinderling et al. 2007). It is evident that the total functional activity of multiple transporters will affect net drug absorbed from the various parts of intestine. A full mechanistic understanding of the intestinal absorption will require: (1) a complete kinetic evaluation of each type of membrane transport, (2) a thorough understanding of local drug concentrations, (3) expression and activity of various transporters, and (4) established, validated and accurately applied *in vitro* models. It is also considered that there is a need to improve the accuracy in predictive software tools, even if it is possible today to obtain comprehensive predictions of the gastrointestinal absorption and bioavailability based on *in silico*, *in vitro*, *in situ* animal data and physiologically based modelling. One contribution would be to perform and generate more high-quality clinical *in vivo* data for intestinal solubility, dissolution, regional permeability, and metabolism.

Drugs are transported by passive diffusion and/or carrier-mediated transport across the intestinal membrane in both the absorptive and secretory directions. Transport proteins that might be involved in multidrug resistance mechanisms have been thoroughly investigated in oncology and have been suggested by many to be a major barrier in drug absorption (Karlsson et al. 1993; Fricker et al. 1996; Greiner et al. 1999; Benet and Cummins 2001; Kruijtz et al. 2002a; Tamura et al. 2002; Glavinas et al. 2004; Kunta and Sinko 2004; Collett et al. 2005; Higgins 2007; Schwarz et al. 2007). Multidrug resistance is most frequently due to active transporters that pump a broad spectrum of chemically distinct, cytotoxic molecules out of cells, including antibiotics, antimalarials, herbicides, and cancer chemotherapeutics in humans. The first and most extensively investigated multidrug transporter is the mammalian P-glycoprotein (Pgp; MDR1; ABCB1), which was identified more than 30 years ago (Dano 1973; Juliano and Ling 1976; Higgins 2007). However, clinical success in reducing multidrug resistance in chemotherapy has been modest. The same observation is valid for various oral drug-delivery strategies to change the activity of the intestinal efflux carriers. There are reports that have raised the issue that the quantitative contribution of membrane efflux transport along the small and large intestine may have been overestimated based on transport data from *in vitro* transport models (Sandstrom et al. 1998a, b, 1999b; Chiou et al. 2001; Hochman et al. 2002; Chiu et al. 2003; Lennernas 2003; Lin 2003; Lin and Yamazaki 2003a, b; Tannergren et al. 2003a, b; Hochman et al. 2004; Lindahl et al. 2004; Petri et al. 2004; Meijer and Lennernas 2005; Cao et al. 2006). The transporter-based drug-drug and/or food-drug interaction(s) have moved the transporter issue further into focus by a recent published US Food and Drug Administration (USFDA) guidance document (Huang et al. 2007). There are several obvious reasons to discuss and evaluate how the efflux by epithelial cells in small and large intestine contribute to pharmacokinetic processes such as absorption, bioavailability, and elimination.

The purpose of this review is to summarize and discuss the reported human intestinal permeability studies. The human permeability data are based on direct *in vivo* determinations in the human gastrointestinal tract with single-pass perfusion systems. The focus of this attention is particularly justified as the availability of directly determined *in vivo* permeability data in the literature are limited, but crucial for the comparison and validation of intestinal *in vitro* and preclinical permeability models as well to form the basis of physiologically based pharmacokinetic modelling. In addition, recent data obtained from a recently developed perfusion model in domestic pigs will be discussed.

Theoretical considerations and data analysis of intestinal transport when applying single-pass intestinal perfusion models

Bioavailability (F) is the most useful pharmacokinetic variable for characterizing the fraction of a dose that reaches the systemic circulation in an unchanged (active) form, and relates it to pharmacological data and safety for oral pharmaceutical products. In a regulatory context, bioavailability is also understood to be the extent and rate at which an active substance or its active moiety is delivered from a pharmaceutical dosage form and becomes available in the general circulation. Bioavailability is mainly dependent on three general but rather complex serial processes; the fraction dose absorbed (f_a), the first-pass extraction of the drug in the gut wall (E_G) and the first-pass extraction of the drug in the liver (E_H) (Equation 1)

(Dressman et al. 1985; Sinko et al. 1991; Amidon et al. 1995; Wu et al. 1995; Lindahl et al. 1996; Cummins et al. 2003; Wu and Benet 2005):

$$F = f_a \cdot (1 - E_G) \cdot (1 - E_H). \quad (1)$$

Different factors might affect the fraction of the dose absorbed (i.e. f_a) and gut wall metabolism (i.e. E_G) of drugs. In general, these can be divided into three broad categories:

- Pharmaceutical factors, such as the choice of excipients and method of manufacturing.
- Physicochemical factors of the drug molecule itself, such as solubility and crystal form.
- Physiological, genetic, biochemical, and pathophysiological factors in the intestine.

f_a is the fraction of the dose transported (absorbed) across the apical cell membrane into the cellular space of the enterocyte according to scientific and regulatory definitions (Amidon et al. 1995; Wu et al. 1995; Lindahl et al. 1996; Cummins et al. 2003; Wu and Benet 2005). Accordingly, the rate (mass/time) and extent of drug absorption ($f_a = \text{mass}/\text{dose}$) from the intestinal lumen *in vivo* are influenced by: dose/dissolution rate ratio, chemical degradation or metabolism in the lumen, luminal complex binding, intestinal transit, and effective permeability (P_{eff}) across the intestinal mucosa. The fraction of the dose absorbed ($f_a = M(t)/\text{dose}$), i.e. the fraction of drug that is lost from the intestinal lumen, assuming no luminal reactions, at any time t is:

$$f_a = \frac{M(t)}{\text{dose}} = \frac{1}{\text{dose}} \int_0^t \int_A P_{\text{eff}} \cdot C_{\text{lumen}} dA' dt', \quad (2)$$

where A is the available intestinal surface area; P_{eff} is the average value of the effective intestinal permeability along the intestinal region where absorption occurs; and C_{lumen} is the free reference concentration of the drug in the intestinal lumen (Amidon et al., 1995). From Equation (2) it is clear that the P_{eff} and the dissolved and free drug concentrations are the key variables controlling the overall rate and extent of absorption, and the equation is thus applicable regardless of the transport mechanism (Amidon et al. 1995; Lennernäs et al. 1995a; Lennernäs 1998; Bourdet et al. 2006).

Once the drug has reached the intracellular site in the enterocyte, it might be metabolized by the cytochrome P450s, predominantly CYP3A4, and may be substrates for other enzymatic processes such as phase II conjugation to glucuronides, sulphates and/or glutathione conjugates. The enzymatic capacity of the small intestine to metabolize drugs can be expressed in pharmacokinetic terms as the extraction ratio of the intestine (E_G) (Sandstrom et al. 1999b; Benet and Cummins 2001; Cummins et al. 2003; Tannergren et al. 2004; Pinto et al. 2005). It is important to realize that CYP3A4 is not expressed in the colon (Nakamura et al. 2002; Thorn et al. 2005; Englund et al. 2006, 2007; Berggren et al. 2007). The heterogeneity of CYP3A4-mediated gut wall metabolism is crucial to consider, especially for the assessment of interactions with other drugs and/or diet as well as development of extended release dosage forms. Importantly, drug metabolism by colonic microflora in the lumen may sometimes reduce the fraction absorbed (f_a) in the colon, particularly for certain drugs given in extended release dosage forms and substrates to hydrolytic and other reductive reactions (Goldin 1990). The fraction that escapes metabolism in the small intestine ($1 - E_G$) may undergo additional metabolism and/or biliary secretion in the liver (E_H) before reaching the systemic circulation. It is important to consider that both a drug and a metabolite (most often a conjugate) do not necessarily have to be eliminated if they are secreted into the intestine since they can undergo enterohepatic circulation. E_H is dependent on blood flow (Q_h), protein binding (f_u), blood/plasma

ratio (C_b/C_p) and intrinsic clearance of enzymes and/or transporters (CL_{int}) and is often based on the well-stirred assumption as shown in Equation (3) (Wu et al. 1995; Rostami-Hodjegan and Tucker 2007; Pang and Rowland 1977a, b, c):

$$E_h = \frac{CL}{Q_h \bullet (C_b/C_p)}, \quad (3)$$

where the intravenous clearance (CL) assumes no extrahepatic metabolism or renal clearance. Recently, it has also been recognized that membrane transport into as well as from hepatocytes must be included in models that predict and explain liver extraction (Takada et al. 2004; Enokizono et al. 2007; Ieiri et al. 2007; Rostami-Hodjegan and Tucker 2007; Soars et al. 2007a, b). These membrane transporters may influence uptake into hepatocytes and canalicular transport into bile. These processes are expected to play a crucial *in vivo* role in the local concentration–time profile in the vicinity of the intracellular enzyme in the liver (Bergman et al. 2006; Petri et al. 2006b; Ghibellini et al. 2007; Leslie et al. 2007).

The permeability of a drug relates to the transport rate across the intestinal mucosa (rate-limiting barrier) and can be applied regardless of transport mechanism. In the intestine the enterocyte is the most common cell type (>90%) in the small intestinal barrier, which also contains a significant number of lymphocytes, mast cells, and macrophages. The intestinal P_{eff} for passive transcellular diffusion is considered to reflect the diffusion across the complex apical membrane into the cytosol, which is situated close to the cytoplasmic leaflet of the apical enterocyte membrane. In addition, it has been speculated that the exofacial leaflet is responsible for the low permeability of the apical membrane (Lande et al. 1995; Lennernas et al. 1995a). Consequently, intestinal single-pass perfusion models that measure the disappearance of the drug from the perfused segment directly describe its quantitative uptake into epithelial cells (Lennernas et al. 1992). Drug transport across the membrane of the perfused jejunal region (i.e. absorption) is the difference between the mass entering and leaving the intestinal segment in Equation (4):

$$\frac{dM}{dt} = Q_{in}C_{in} - Q_{out}C_{out} = Q(C_{in} - C_{out}), \quad (4)$$

where C_{in} and C_{out} are the input and output drug concentrations, respectively; and Q is the flow through the tube. The mass balance relationship has previously been proposed to describe the transport rate of the drug across the whole mucosal barrier (absorbed mass) according to Fick's first law in Equation (5) (Flynn et al. 1974; Amidon et al. 1995; Lennernas 1995):

$$dM/dt = A \bullet P_{eff} (C_{ref}^{lumen} - C_{ref}^{blood}) \quad (5)$$

where A is the surface area of the membrane; P_{eff} is the effective permeability and the reference concentrations, C_{ref}^{lumen} and C_{ref}^{blood} , are on the opposite sides of the intestinal mucosa.

Based on the discussion in the previous section in which the apical cell membrane was considered to be rate limiting for absorption, the following relationship might be derived (Fagerholm et al. 1995, 1999; Lande et al. 1995; Lennernas 1997):

$$\frac{dM}{dt} = A \bullet P_{eff} (C_{ref}^{lumen} - C_{ref}^{cytosol}). \quad (6)$$

However, since it is not possible to measure the concentration of a drug in the cytosol of enterocytes or in the vena porta in humans, peripheral venous blood was used as the *in vivo*

reference on the receiver side. For example, it has been shown that for antipyrine the portal and peripheral plasma concentrations are about 100-fold less than the luminal concentration (Lennernäs et al. 1992; Petri et al. 2006b). These data indicate that it is valid to assume sink conditions as antipyrine is a drug with a small volume of distribution — approximately 40 litres in a 70-kg person — and is assumed to be evenly distributed in the body (Johnson and Amidon 1988; Lennernäs et al. 1992; Lennernäs 1998; Petri et al. 2006b). In addition, a similar value for the liver extraction of fluvastatin — a CYP2C9 substrate — has been reported after jejunal perfusion and intravenous administration, which supports the mass balance concept established between disappearance and appearance transport rates during a single-pass perfusion experiment utilizing the Loc-I-Gut[®] technique (Lindahl et al. 1996).

P_{eff} and other variables were calculated from the steady-state level in the perfusate leaving the jejunal segment. The compounds of interest in the perfusate within the intestinal segment achieved equilibrium when the concentrations of the solute and the ¹⁴C-polyethylene glycol (PEG) 4000 in the outlet perfusate reached a plateau (after 50–60 min). Based on a residence time distribution analysis, it has been reported that the hydrodynamics within the perfused jejunal segment were best described by a well-mixed model (Lennernäs et al. 1992, 1997b). P_{eff} was calculated using the outlet concentration as a reference concentration according to Equation (7):

$$P_{\text{eff}} = Q_{\text{in}} \bullet \frac{(C_{\text{in}} - C_{\text{out}})}{(C_{\text{out}} \cdot 2\pi Rl)}, \quad (7)$$

where Q_{in} is the input perfusate rate; C_{in} and C_{out} are the input and output perfusate concentrations of the drug, respectively; R is the radius ($=1.75$ cm); and l is the length of the jejunal segment (10 cm). Q_{in} is the perfusion flow rate entering the jejunal segment. In all experiments, the stability and adsorption of the drugs were carefully assessed (at 37°C for 180 min) and no degradation and adsorption were detected for any of the drugs investigated. In the dog and pig models, permeability was calculated using the well-stirred model (Equation 7) since it is assumed that the hydrodynamics are similar to the corresponding model in the human intestine (Petri et al. 2006b). The cylinder surface area is considered to be the accurate area to use for high-permeability drugs since they are transported from the tip of the intestinal villus. For low-permeability drugs, the actual absorbing surface area *in vivo* is probably larger than the cylinder area. This is not possible to measure and since studies are compared with each other this is no limitation. However, it seems that the *in vivo* scaled permeability values for low-permeability drugs predict the *in vivo* absorption as previously reported (Lennernäs 1997, 1998; Lennernäs et al. 1997a). It should also be remembered that all cell culture models are based on operative permeability calculated from the area of a monolayer.

Steady-state effective intestinal permeability (P_{eff} , cm s^{-1}) in the human intestine with an open or a semi-open single-pass perfusion model is usually calculated using the parallel-tube model, since they are perfused from an entrance at one end of the intestinal segment to exit at the other end (equation 8):

$$P_{\text{eff}} = \frac{(-Q_{\text{in}} \bullet \ln(C_{\text{out}}/C_{\text{in}}))}{A}, \quad (8)$$

where C_{in} and C_{out} are the concentrations of each compound, corrected for input and output fluid transport, respectively; Q_{in} is the single-pass perfusion flow rate; and A is the mass transfer surface area within the intestinal segment, which is assumed to be a cylinder with a height of 10 cm and a radius of 1.75 cm.

Human jejunal permeability values (P_{eff}) are determined in the region with the highest expression of carrier proteins in the intestine and largest surface area. These P_{eff} values are often based on multiple parallel transport processes such as passive diffusion, carrier-mediated absorption, and efflux. Site-specific P_{eff} cannot reflect the permeability along the intestinal tract, but they are useful for approximations of the fraction of the oral dose absorbed. It seems like drugs with a jejunal $P_{\text{eff}} > 1.5 \times 10^{-4} \text{ cm s}^{-1}$ will be completely absorbed no matter which transport mechanism(s) are utilized. Furthermore, metoprolol has a jejunal $P_{\text{eff}} = 1.34 \times 10^{-4} \text{ cm s}^{-1}$ and are absorbed mainly by passive diffusion and will also be completely absorbed throughout the human colon. Many drugs, despite the fact that they are subjected to significant efflux, have a rapid and extensive absorption by passive transcellular diffusion, which will result in a high fraction of the oral dose absorbed (i.e. >85%). The determined jejunal P_{eff} for drugs transported mainly by solute carrier class (SLC) carrier in the absorptive direction (such as peptide and amino acid transported drugs) will be good estimates for the prediction of fraction of the dose absorbed, since such a drug will only be absorbed in the small intestine (this is the region where these absorptive SLC carriers are expressed and has a negligible passive absorption in the colon). Accordingly there is a need to investigate regional permeability, especially in human colon. This would increase knowledge about the important mechanisms, provide high-quality data with the aim of developing and validating rapid and more reliable *in vitro* intestinal models, and form the basis for a reliable physiologically based modelling of oral drug absorption.

Brief overview about how to determine the effective human jejunal permeability for drugs

Intestinal permeability (P_{eff}), together with solubility, dissolution rate and gastrointestinal transit, is one of the key biopharmaceutical variables that determine the *in vivo* rate and extent of intestinal drug absorption in humans and animals (Sinko et al. 1991; Lennernas et al. 1992; Amidon et al. 1995). The physicochemical properties of the drug and the complex physiological and biochemical conditions of the gastrointestinal tract will also strongly affect intestinal P_{eff} *in vivo*. Variations in gastrointestinal physiology might affect the P_{eff} differently, depending on the transport mechanism(s) involved (Chadwick et al. 1977a, b; Corrigan 1997; Lennernas 1998; Ungell et al. 1998; Winiwarter et al. 1998, 2003; Davis and Wilding 2001). It is important to consider that intestinal P_{eff} may depend on multiple, parallel transport processes such as passive transcellular diffusion, carrier-mediated absorption, and carrier-mediated efflux.

Direct measurements of intestinal absorption, secretion, and metabolism of drugs in humans are possible by regional intestinal perfusion techniques. In general, four different clinical tools have been employed in the small intestine (Modigliani et al. 1973a, b; Rambaud et al. 1973; Lennernas et al. 1992, 1994; Lennernas 1998; Von Richter et al. 2001; Drescher et al. 2003; Igel et al. 2007):

- Triple-lumen tube including a mixing segment.
- Multilumen tube with a proximal occluding balloon.
- Multilumen tube (Loc-I-Gut) with two balloons occluding a 10-cm-long intestinal segment (Figure 1).
- Two 20-cm adjacent jejunal segments were isolated with the multilumen perfusion catheter.

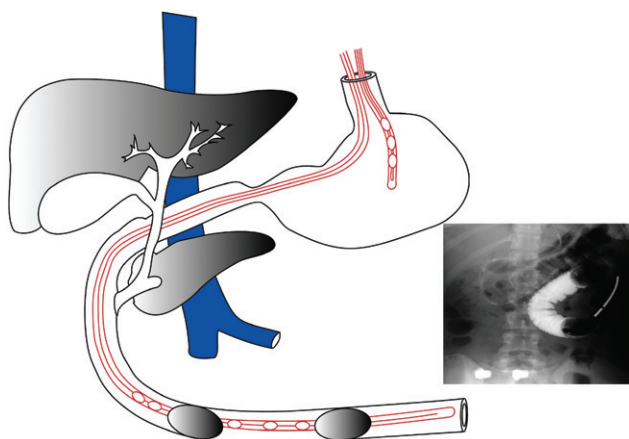


Figure 1. Loc-I-Gut is a perfusion technique for the proximal region of the human jejunum. The multichannel tube is 175 cm long and is made of polyvinyl chloride with an external diameter of 5.3 mm. It contains six channels and is provided distally with two 40-mm-long elongated latex balloons, placed 10 cm apart, and each separately connected to one of the smaller channels. The two wider channels in the centre of the tube are for infusion and aspiration of perfusate. The two remaining peripheral smaller channels are used for administration of marker substances and/or for drainage. A tungsten weight is attached to the distal end of the tube in order to facilitate passage of the tube into the jejunum. The balloons are filled with air when the proximal balloon has passed the ligament of Treitz. Gastric suction is obtained by a separate tube. ^{14}C -PEG 4000 is used as a volume marker to detect water flux across the intestinal barrier. The inset shows the tube in place within the human body determined with traditional X-ray and barium sulphate perfusion of the small intestinal segment.

The advantages and disadvantages of the various intestinal perfusion techniques are discussed elsewhere (Lennernäs 1998). Clinical studies of effective intestinal permeability (P_{eff}), secretion and metabolism of various compounds, such as drugs, environmental pollutants and nutrients are rarely performed *in vivo* in humans, even if experimental techniques are available (Figure 1). Direct measurements of compound transport and metabolism in the mesenteric vein and portal vein in humans are not possible for obvious reasons. Perfusion techniques do, however, offer great possibilities of measuring various intestinal processes. Over the past 70 years different *in vivo* intestinal perfusion techniques have been developed and the importance of such *in vivo* work has been clearly demonstrated (Modigliani et al. 1973a, b; Rambaud et al. 1973; Pfeiffer et al. 1990; Lennernäs et al. 1992, 1994; Lennernäs 1998; Von Richter et al. 2001; Drescher et al. 2003; Igel et al. 2007). The major advantages to using P_{eff} as the absorption parameter are that: (1) it is possible to measure regardless of the transport mechanism(s) across the intestinal mucosa, (2) predicts the f_a , and (3) can be used to assess *in vitro*–*in vivo* correlations which validates the use of different intestinal absorption models commonly applied in discovery and preclinical development (Lennernäs 1998). It is important to recognize that such *in vivo* studies of intestinal absorption and function provide a fully integrated response in humans, with a relevant combination of genetic, biochemical, physiological, pathological, and environmental factors. A good correlation has been established between *in vivo* determined P_{eff} and historical data on f_a for a large number of structurally diverse drugs (Table I and Figure 2). However, one criticism that has been levelled on perfusion techniques in general is that they may disturb the gastrointestinal physiology, especially the motility patterns (Davis and Wilding 2001). In a study reported by Read et al. (1983), the effect of a tube that reached the terminal ileum on gastrointestinal physiology was investigated by monitoring the

Table I. Biopharmaceutics classification system (BCS) classification of 28 drugs based on human effective permeability (P_{eff}) and dose number. Each P_{eff} value was determined *in vivo* in the proximal jejunum in humans with a single-pass approach at pH 6.5 (phosphate buffer) and under isotonic conditions. Twenty-four of the drugs were evaluated at Uppsala University, Sweden, and five were evaluated at the University of Michigan, USA.

Drug	Human <i>in vivo</i> permeability ($\times 10^4 \text{ cm s}^{-1}$)	Dose number*	BCS Class	f_a (%)	Laboratory
α -Methyldopa	0.10	0.1	III	55–65	UU
Amiloride	1.6	0.4–0.8	I	80–90	UU
Amoxicillin [#]	0.30	0.9	III	45–75	UU
Antipyrine	5.60	0.20	I	100	UU
Atenolol	0.20	0.02	III	50–60	UU
Carbamazepine	4.30	80	II	>90	UU
Cephalexin	1.56	2	II	>90	UM
Cimetidine	0.26	3	III	75	UU
Cyclosporine	1.61	350	II	>90	UM
Desipramine HCl	4.50	<0.03	I	100	UU
Enalapril maleate	1.57	0.003	(I)**	65	UM
Enalaprilat	0.20	0.003	III	8	UU
Fexofenadine	0.07	0.32	III	5–10	UU
Fluvastatin sodium	2.40	<0.9	I	95	UU
Furosemide	0.05	30	IV	40–60	UU
Hydrochlorothiazide	0.04	0.2	III	55	UU
Isotretinoin	0.99	>20	II	90	UU
Inogatran	0.03	<0.001	III	5–10	UU
Ketoprofen	8.70	0.2	I	100	UU
L-dopa	3.40	1.0	(I)***	100	UU
Lisinopril	0.33	0.002	III	35	UU
Losartan	1.15	0.004	III	100	UU
Metoprolol	1.34	0.0004	I	95	UU
Naproxen	8.50	0.06	I	100	UU
Piroxicam	6.65	2.5	II	100	UM
Propanolol	2.91	0.01	I	100	UU
Ranitidine	0.27	0.01	III	50–60	UM
Terbutaline	0.30	0.01	III	40–50	UU
Valacyclovir	1.66	0.02	I***	>80	UM
R-verapamil	6.80	0.004	I	100	UU
S-verapamil	6.80	0.004	I	100	UU

*Human P_{eff} was determined at a concentration based on the most common clinical dose dissolved in 250 ml. For low solubility concentration, the highest possible drug concentrations were applied.

**Dose number = dose/ $V_0/C_{s \text{ min}}$ (highest dose strength/initial gastric volume (250 ml)) min^{-1} imum solubility.

***High permeability due to carrier-mediated absorption, currently not included in BCS Class I.

UU, Uppsala University, Sweden; UM, University of Michigan, USA.

[#]75% at 500 mg 45% at 3000 mg.

Data are from: Lennernas et al. (1992; 1993; 1994, 1997a, 2002b); Fagerholm et al. (1995; 1996, 1997, 1999); Lindahl et al. (1996); Lennernas (1997, 1998); Soderholm et al. (1997); Takamatsu et al. (1997, 2001); Sandstrom et al. (1998b, 1999b); Winiwarer et al. (1998, 2003); Sun et al. (2002); Chiu et al. (2003); Petri et al. (2003, 2006b); and Tannergren et al. (2003a, b; 2004).

gastrointestinal passage of a solid meal with gamma-scintigraphy. Read et al. reported that the gastric emptying half-life changed from 1.2 ± 0.32 to 1.5 ± 0.35 h, and the small intestinal transit time decreased from 3.6 ± 1.33 to 2.0 ± 0.99 h (mean $\pm$ standard deviation (SD)). Thus, the effects of the tube on gastric emptying are minimal and do not question the pharmaceutical relevance of drug absorption data collected using these perfusion methods. Further support for this conclusion is reported by Naslund et al. (2000) who clearly showed that there was no difference in gastric emptying between the following

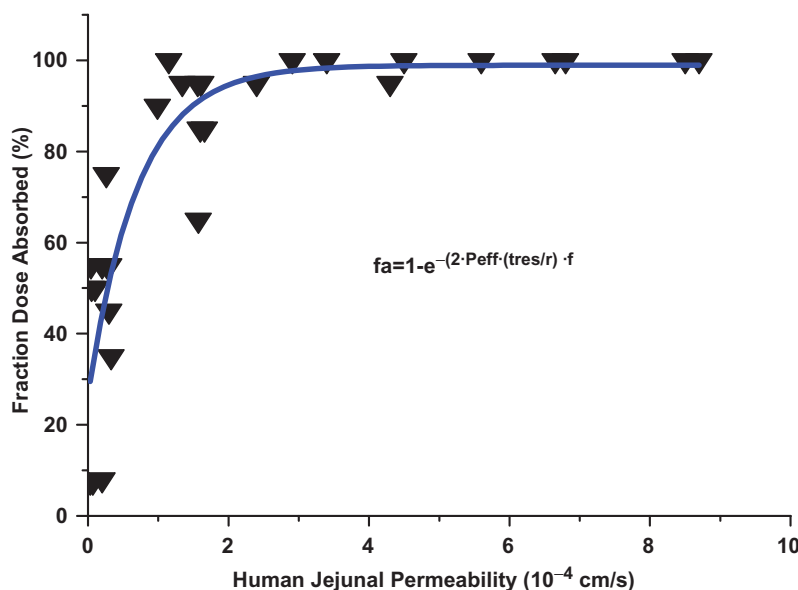


Figure 2. Human *in vivo* permeability values (P_{eff}) were determined by the use of a single-pass perfusion technique (Loc-I-Gut) in human jejunum. These human P_{eff} have been correlated to the fraction dose absorbed (f_a) of oral doses for a large number of drugs from different pharmacological classes which consequently represent structural diversity. Human *in vivo* jejunal permeability values for 42 compounds (31 drugs) were determined over a period of 18 years by applying this clinical technique and are presented in Tables I and II (Lennernäs et al. 1992, 1993, 1994, 1997a, 2002b; Fagerholm et al. 1995, 1996, 1997, 1999; Lindahl et al. 1996; Lennernäs 1997, 1998; Soderholm et al. 1997; Takamatsu et al. 1997, 2001; Sandstrom et al. 1998b, 1999b; Winiwarer et al. 1998, 2003; Sun et al. 2002; Chiu et al. 2003; Petri et al. 2003, 2006b; Tannergren et al. 2003a, b, 2004).

three methods: scintigraphic, oral dosing of paracetamol tracer, and subsequent plasma sampling, and PEG dilution methods using intubation tubes.

Validated methods have been used for regional single-pass perfusions of the proximal jejunum and distal rectum *in vivo* in humans on separate occasions. The small intestinal tube has been extensively used to examine the effective permeability of various compounds (Figure 1) (Lennernäs et al. 1992, 1994; Lennernäs 1998; Tannergren et al. 2003a, b). The jejunum is the major absorbing region for drugs and nutrients in most mammals. It also has the largest surface area and is the site of the most active carrier-mediated transport in the gut (Chadwick et al. 1977a, b; Lennernäs et al. 1992; Collett et al. 1997; Ungell et al. 1998; Hilgendorf et al. 2007). Human *in vivo* jejunal permeability values for 42 compounds (31 drugs) have been determined over a period of 18 years by applying this clinical technique. They are presented in Tables I and II (Lennernäs et al. 1992, 1993, 1994, 1997a, 2002b; Fagerholm et al. 1995, 1996, 1997, 1999; Lindahl et al. 1996; Lennernäs 1997, 1998; Soderholm et al. 1997; Takamatsu et al. 1997, 2001; Sandstrom et al. 1998b, 1999b; Winiwarer et al. 1998, 2003; Sun et al. 2002; Chiu et al. 2003; Petri et al. 2003, 2006b; Tannergren et al. 2003a, b, 2004). Of these 42 compounds, the majority of the permeability values have been determined in one laboratory at Uppsala University. The jejunal P_{eff} have been determined at buffer pH 6.5 and the highest drug concentration possible in relation to safety and solubility properties of each drug.

The human *in vivo* jejunal P_{eff} (at pH 6.5) for fexofenadine, furosemide, hydrochlorthiazide, and inogatrán are 0.07×10^{-4} , 0.05×10^{-4} , 0.04×10^{-4} , and $0.03 \times 10^{-4} \text{ cm s}^{-1}$,

Table II. Effective permeability (P_{eff}) of ten compounds determined by *in vivo* single-pass perfusion of the proximal jejunum in humans. Six compounds were evaluated at Uppsala University, Sweden, one was evaluated at the University of Michigan, USA, and three were evaluated at the University of Linköping, Sweden. Human *in vivo* jejunal permeability values for 42 compounds (31 drugs) were determined over a period of 18 years by applying this clinical technique and are presented in Tables I and II.

Drug	Human <i>in vivo</i> permeability ($\times 10^{-4}$ cm s $^{-1}$)	BCS Class	Laboratory
L-leucine	6.20	I	UU
L-phenylalanine	4.08	I	UM
D-glucose	10.0	I	UU
Sulforaphane	18.7	II	UU
Creatinine	0.29	III	UU
Urea	1.4	I	UU
Water (D $_2$ O)	1.4	I	UU
PEG 546	1.6	I	UL
PEG 810	0.9	III	UL
PEG 942	0.95	III	UL

*Human P_{eff} was determined at a concentration based on most common clinical dose dissolved in 250 ml. For low solubility concentration, the highest possible drug concentrations were applied.

UU, Uppsala University, Sweden; UM, University of Michigan, USA; UL, University of Linköping, Sweden.

Data are from: Lennernas et al. (1992, 1993, 1994, 1997a, 2002b); Fagerholm et al. (1995, 1996, 1997, 1999); Lindahl et al. (1996); Lennernas (1997, 1998); Soderholm et al. (1997); Takamatsu et al. (1997, 2001); Sandstrom et al. (1998b, 1999b); Winiwarter et al. (1998, 2003); Sun et al. (2002); Chiu et al. (2003); Petri et al. (2003, 2006b); Tannergren et al. (2003a, b, 2004).

respectively (Table I) (Lennernas et al. 1997a; Winiwarter et al. 1998; Tannergren et al. 2003a). The fraction dose absorbed for each drug is 10–20, 40–60, 55, and 5–10%, respectively (Table I). These drugs have hydrophilic properties and their passive transcellular diffusion is expected to be low. They are also too large to be considered to have a significant paracellular uptake, which also is indicated by the low and incomplete fraction dose absorbed. A potential explanation for the relative high interindividual variability seen in the permeability estimate is that there is a small difference between inlet and outlet concentration (see Equation 7). For the two diuretics there was also a strong induction of fluid flux into the segment that might have affected the determination of the permeability value. Nevertheless, the determined human *in vivo* P_{eff} has been successfully classified for fexofenadine, furosemide hydrochloride, and inogatran as low-permeability compounds with a high interindividual variability in absorption.

During the past decade, significant efforts have been made to investigate the effect of cellular carrier efflux on the pharmacokinetics of drugs and their metabolites (Cummins et al. 2003; Petri et al. 2004; Witthoft et al. 2006; Kalvass and Pollack 2007). Most of the understanding of the mechanisms behind intestinal efflux is derived from cell culture studies and gene knock-out animals (Karlsson et al. 1993; Fricker et al. 1996; Engman et al. 2001; Stephens et al. 2001, 2002a, b; Collett et al. 2005). Unfortunately, direct kinetic comparisons of transport variables under the complicated circumstances of *in vivo* settings in humans are rare and the *in vivo* relevance for intestinal efflux might have been significantly over-interpreted (Nilsson et al. 1994; Lindahl et al. 1996; Lennernas 1998; Sandstrom et al. 1999b; Chiu et al. 2003; Tannergren et al. 2003a, b, 2004; Cao et al. 2006; Igel et al. 2007). The *in vivo* pharmacokinetic evaluation of the quantitative importance of drug transporters suffers from the lack of good *in vivo* probe molecules and specific transport inhibitors. There is also a need to use other approaches, such as *in vivo* perfusion techniques and molecular imaging techniques besides plasma concentration measurement since the resolution of the data might be insufficient to define the transport mechanisms that apply

(Tucker et al. 2001). This is clearly a very important issue, as data on drug–drug interactions are now prioritized clinical and regulatory demands (Huang et al. 2007). Intestinal perfusion techniques (such as Loc-I-Gut), for instance, might enable significantly more direct *in vivo* investigations of intestinal absorption, efflux, and biliary secretion, without the influence of dissolution, gastrointestinal motility, transit, pH changes, and metabolism (Sandstrom et al. 1998b; Chiu et al. 2003; Tannergren et al. 2003a, b, 2004). A single-pass perfusion of the intestine might be viewed as the maximal risk for a drug–drug interaction since both the substrate and inhibitor concentrations are controlled within the single-pass perfused segment. Research based on this single-pass perfusion technique has produced a large database consisting of *in vivo* drug permeability and biliary secretion data that might be used to compare directly permeability data among various animal, cell culture models, and perfusion bioreactor models in the future (Fagerholm et al. 1996; Lennernas et al. 1997a; Takamatsu et al. 2001; Sun et al. 2002; Kim et al. 2007).

Unstirred water layer

Previously, the unstirred water layer adjacent to the intestinal lining was considered to be the rate-limiting step for intestinal P_{eff} of high-permeability compounds. Hydrodynamics within the intestinal segment can best be described by a well-stirred model and P_{eff} was calculated according to Equation (7) (Lennernas et al. 1997b). The reciprocal of P_{eff} ($1/P_{\text{eff}}$) is an estimate for the total effective resistance (R_{eff}) to drug absorption, and is a combination of two diffusional resistances in series, namely the unstirred water layer ($1/P_{\text{aq}}$) and the cell membrane ($1/P_{\text{m}}$) (Johnson and Amidon 1988):

$$R_{\text{eff}} = \frac{1}{P_{\text{eff}}} = \frac{1}{P_{\text{aq}}} + \frac{1}{P_{\text{m}}} \quad (9)$$

According to the widely used assumption that rapidly absorbed solutes have negligible resistance to passage through the cell membrane ($P_{\text{m}} \gg P_{\text{aq}}$) (Winne 1978a, b), Equation (9) can be simplified to:

$$P_{\text{eff}} \approx P_{\text{aq}} \quad (10)$$

The effective unstirred water layer thickness (∂) can then be estimated by the use of the following equation (Winne 1978a, b):

$$\partial \approx \frac{D}{P_{\text{eff}}}, \quad (11)$$

where D is the diffusion coefficients of antipyrine ($0.91 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$) and D-glucose ($0.80 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$) in buffer solution or water at 37°C (Lennernas 1998). The estimation of P_{eff} and ∂ from perfusion experiments is based upon several assumptions (Table III), e.g. (1) the intestine has an effective absorptive area of a smooth cylinder, (2) the diffusion rates obtained in plain saline buffer equal that in the unstirred water layer, (3) the intestinal segment is totally filled with perfusate, and (4) the enterocyte membrane is considered to be a negligible barrier to intestinal absorption of high P_{eff} solutes. The results presented in Table III are given as median and percentiles (25 and 75%) due to the large interindividual variability in the human permeability data. Several *in vivo* studies report that the thickness of this unstirred water layer is significantly thinner ($<40\text{--}150 \mu\text{m}$) than was previously assumed (approximately $700\text{--}1000 \mu\text{m}$), since there is an instantaneous and intense mixing of

intestinal fluids by the forceful small intestinal motility. For instance, Fagerholm and Lennernäs (1995) observed no significant changes in estimated P_{eff} of two high-permeability (human *in vivo* jejunal $P_{\text{eff}} > 2 \times 10^{-4} \text{ cm s}^{-1}$) compounds (absorbed with active transport and passive diffusion), D-glucose ($10 \times 10^{-4} \text{ cm s}^{-1}$) and antipyrine ($4 \times 10^{-4} \text{ cm s}^{-1}$), or unstirred water layer thickness over a fourfold range of single-pass perfusion rates ($1.5\text{--}6.0 \text{ ml min}^{-1}$) when using the Loc-I-Gut double balloon technique in humans (Figure 3 and Table III) (Fagerholm and Lennernäs 1995). Therefore, it is currently accepted that the epithelial membrane controls the transport intestinal rate for both low- and high-permeability compounds regardless of the transport mechanism *in vivo*.

Table III. Effective human *in vivo* jejunal permeability (P_{eff}) and effective human jejunal unstirred water layer thickness (δ) for antipyrine (0.5–10.5 mM) and D-glucose (10 mM). Permeability values were determined by single-pass perfusion. Antipyrine is a very weak base ($\text{pK}_{\text{a}} = 1.44$) and is not subjected to any apical efflux by enterocytes. It is a small and sufficiently hydrophobic drug ($\log P$, $\log D = 0.6$) to be absorbed with passive transcellular diffusion. D-glucose is absorbed by an Na^{+} -dependent transporter for D-glucose (SGLT1).

Perfusion flow rate (ml min^{-1})	$P_{\text{eff}} (\times 10^{-4} \text{ cm s}^{-1})$		$\delta (\mu\text{m})$		Number of antipyrine/D-glucose
	Antipyrine	D-glucose	Antipyrine	D-glucose	
1.5	3.5 (–)	–	274 (–)	–	2/–
2.0	3.3 (2.4–5.5)	6.2 (3.5–14)	280 (170–396)	128 (57–231)	16/15
3.0	5.1 (3.4–9.0)	11 (7.3–18)	177 (101–268)	74 (43–110)	49/30
6.0	6.0 (3.1–18)	9.3 (3.4–44)	152 (68–402)	96 (41–303)	8/8
Total median	4.8	9.6	188	83	75/53

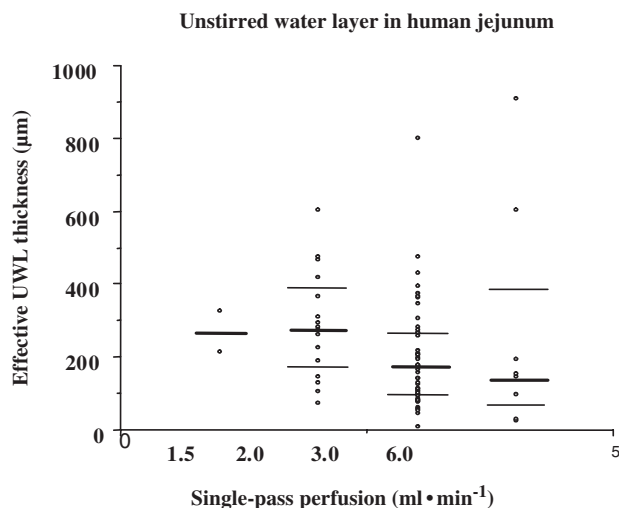


Figure 3. Individually unstirred water layer thickness data (δ) calculated from P_{eff} estimates of antipyrine (left) and D-glucose (right) at different perfusion flow rates in the human jejunum: (—) and (– – –) represents the median and 25% and 75% percentiles, respectively. Data are from Fagerholm and Lennernäs (1995b).

Paracellular absorption

Paracellular diffusion is now considered as a minor intestinal transport route *in vivo* for drug molecules larger than molecular weight (MW) 250–300 (Lennernäs et al. 1994; Nilsson et al. 1994; Fagerholm et al. 1995, 1999; Lennernäs 1995, 1997, 1998; Collett et al. 1997; He et al. 1998). The enterocytes are sealed together by negatively charged tight junctions, and the intracellular space formed is considered to be the paracellular route. The available surface area for paracellular intestinal absorption has been estimated to be about 0.01% of the total surface area of the small intestine. The quantitative importance of the paracellular route for the macroscopic intestinal absorption of hydrophilic compounds *in vivo* is not yet fully clear. There have been several *in vitro* investigations demonstrating that the paracellular route is important for the intestinal absorption of various hydrophilic compounds. However, *in vivo* studies have suggested that this route makes only a minor contribution to the overall intestinal absorption of drugs (Lennernäs et al. 1994; Lennernäs 1995; Fagerholm et al. 1999). The quantitative importance of the paracellular route for the macroscopic intestinal absorption of hydrophilic compounds *in vivo* has been investigated by using the jejunal and rectal perfusion techniques. For instance, water transport during solvent drag is largely paracellular, and based on this assumption it has been suggested that compounds with a MW of over approximately 200 Da (radius >4.0 Å) are too large for the limited intercellular space between the enterocytes, and therefore not sensitive to solvent drag in terms of quantitative absorption (Lennernäs et al. 1994; Lennernäs 1995; Fagerholm et al. 1999). This hypothesis is supported by observations that in humans small hydrophilic compounds such as urea (MW = 60 Da, molecular radius = 2.6 Å) and creatinine (MW = 113 Da, molecular size = 7.2–8.0 Å) are affected by solvent drag in humans, whereas other hydrophilic compounds with a MW >180 Da, such as D-glucose (MW = 180 Da), antipyrine (MW = 188 Da), L-dopa (MW = 197 Da), terbutaline (MW = 225 Da), atenolol (MW = 266 Da), and enalaprilat (MW = 348 Da) are not (Lennernäs et al. 1994; Nilsson et al. 1994; Fagerholm et al. 1995, 1999; Lennernäs 1995). Further support for passive transcellular transport of fairly hydrophilic drugs are the observations that atenolol ($\log D_{6.5} < -2$, polar surface area (PSA) = 88 Å^2 , hydrogen bonding donors (HBD) = 4) inhibits Pgp-mediated efflux and that terbutaline ($\log D_{6.5} < -1.3$, PSA = 76 Å^2 , HBD = 4) is extensively metabolized in the gut wall during first-pass extraction following oral administration (Karlsson et al. 1993; Borgström et al. 1989). Both these compounds have been suggested to be largely absorbed by the paracellular route due to their hydrophilic properties. However, atenolol has been reported to decrease the basal apical transport of celiprolol — a Pgp substrate that does not undergo CYP3A4 metabolism in Caco-2 cells (Karlsson et al. 1993). This transport inhibition is a competitive interaction at the binding sites of the Pgp, which are located within the transmembrane region of the Pgp. Terbutaline undergoes extensive sulphate conjugation after oral administration, which appears to predominately occur in the gut wall. This particular conjugation enzyme is located in the cytosolic fraction of the enterocyte, which indicates that terbutaline has to be transported via the transcellular route to be metabolized, even if it is a hydrophilic drug ($\log D_{6.5} = -1.3$) with low jejunal P_{eff} ($0.3 \times 10^{-4} \text{ cm s}^{-1}$) (Borgström et al. 1989). The human *in vivo* perfusion data together with this discussion on two hydrophilic drugs, atenolol and terbutaline, support the hypothesis that small and fairly hydrophilic drugs are mainly absorbed via the transcellular route if passive diffusion is the predominant intestinal absorption mechanism. In addition, Soergel (1993) suggests that the intestinal mucosa is essentially impermeable to paracellular transport of hexoses, and the paracellular route is unlikely to make a major contribution to the small intestinal absorption of bile acids

(i.e. of MW = 500–600 Da) (Amelsberg et al. 1996). Based on these *in vivo* data and their interpretation, it is still surprising that the regulation of tight junctions and increased paracellular absorption of even large hydrophilic compounds is considered as a possible *in vivo* approach to increase the absorption of even large and low permeability compounds (Tavelin et al. 2003). Taken collectively and based on this drug-delivery principle it is clear that it is extremely difficult to explore novel approaches that regulate the intestinal barrier and at the same time increase drug absorption in a safe and specific manner without any serious side-effects (Lindmark et al. 1997; Lennernas et al. 2002a).

Transcellular absorption

The main intestinal absorption mechanism for drugs *in vivo* is considered to be passive transcellular membrane diffusion with the apical membrane as the rate-limiting step and is consistent with the lipophilic properties that most drugs exhibit (Lande et al. 1995; Lennernas 1995; Skold et al. 2006). For instance, in a pharmacokinetic database of a total of 472 drugs, it been reported that 235 drugs (50%) had a log $P > 2$ and 379 (80%) had log $P > 0$. This is further supported by the fact that the majority (approximately 70%) of the approved drugs are eliminated by metabolism due to their lipophilic nature (Van de Waterbeemd 1998; Smith and van de Waterbeemd 1999). This classification of approved drugs is also in accordance with the recently proposed Biopharmaceutics Drug Disposition Classification System (BDDCS) (Wu and Benet 2005). Even if a drug is a substrate for an intestinal transport protein, passive transcellular diffusion will probably be the main absorption mechanism if the drug has suitable lipophilic physiochemical properties (such as log $P > 0$) (due to unlimited capacity and larger surface area). Almost all high-permeability drugs will also undergo complete metabolism, which further support earlier suggestions that *in vitro* efflux by cell monolayers has significantly been over-interpreted as a barrier for oral drug absorption (Lennernas 1998, 2003; Sandstrom et al. 1998b; Chiou et al. 2001; Lin and Yamazaki 2003a, b; Lindahl et al. 2004; Tannergren et al. 2004; Cao et al. 2006). It has just recently been suggested that metabolism might be added (not substituted) to the present biopharmaceutics classification system (BCS) requirement to facilitate a significant number of marketed drugs being appropriately assigned to BCS Class 1 (Benet et al. 2007). On the other hand, intestinal carrier-mediated membrane transport might dominate for some hydrophilic drugs and polar metabolites if they are substrates for any transporter, since the passive membrane diffusion is then expected to be slow for such compounds. Intestinal efflux has been reported to affect the *in vivo* absorption of drugs such as digoxin, which is a drug with a low solubility and given in a low dose (1 mg) (Greiner et al. 1999; Drescher et al. 2003; Igel et al. 2007). The role of biliary excretion has also been suggested to play a role for the systemic availability and transported mediated drug–drug interactions of digoxin (Angelin et al. 1987; Hedman et al. 1990; 1991a, b, 1992; Hedman 1992; Hedman and Meijer 1998a, b).

It is now well known that the intestinal absorption potential of drugs that are mainly transported by passive diffusion may be predicted from molecular properties, such as polar surface area, hydrogen bonding, log P , log D and molecular weight (Lipinski et al. 2001). Hence, better *in silico* models are required that can predict permeability for chemical structure outside the established chemical space and which can then be used to define the relationship between physicochemical descriptors and permeability (Skold et al. 2006). The values for *in vivo* jejunal human permeability (P_{eff}) (Loc-I-Gut) for 22 chemically and

Table IV. Human effective permeability (P_{eff}) of 21 compounds and a few of their physicochemical properties. Each P_{eff} value was determined *in vivo* in the proximal jejunum in humans with a single-pass approach at pH 6.5 (phosphate buffer) and under isotonic conditions at Uppsala University, Sweden.

Compound	Log P_{eff}	Pred_log P_{eff}	MW	PSA	HBD
1. Amoxicillin	-4.40	-5.63	365.40	154.38	5
2. Antipyrine	-3.35	-2.84	188.23	26.50	0
3. Atenolol	-4.70	-4.63	266.34	88.05	4
4. Carbamazepine	-3.38	-3.56	236.27	41.59	2
5. Creatinine	-4.54	-3.84	113.12	67.31	2
6. Desipramine	-3.35	-3.00	266.39	16.16	1
7. Enalaprilat	-4.70	-4.50	348.40	102.07	3
8. Fluvastatin	-3.62	-4.28	411.47	81.37	3
9. Furosemide	-5.40	-5.03	330.74	124.30	4
10. Glucose	-3.00	-5.19	180.16	113.63	5
11. Hydrochlorothiazide	-5.40	-5.12	297.73	132.6	4
12. Ketoprofen	-3.07	-3.42	254.29	54.23	1
13. L-leucine	-3.21	-4.11	131.18	66.31	3
14. L-dopa	-3.47	-5.10	197.19	105.51	5
15. Methyl dopa	-4.70	-5.06	211.22	102.56	5
16. Metoprolol	-3.92	-3.74	267.37	57.81	2
17. Naproxen	-3.00	-3.35	230.26	48.18	1
18. Piroxicam	-3.21	-4.10	331.35	90.90	2
19. Propranolol	-3.54	-3.53	259.35	39.22	2
20. Terbutaline	-4.52	-4.50	225.29	76.38	4
21. Urea	-3.85	-4.51	60.06	77.50	4

Data are from Winnivarter et al. (1998).

MW, molecular weight (g/mol); PSA, polar surface area ($\text{\AA}^2$); HBD, hydrogen bonding donors.

structurally diverse compounds were correlated in a multivariate data analysis that included both experimentally determined lipophilicity values obtained using the pH-metric technique and calculated molecular descriptors (Table IV) (Winiwarter et al. 1998, 2003). One of the models obtained from this multivariate analysis was used to predict the log P_{eff} values for an external validation set comprised of 34 compounds. The data correlated well with human absorption data for the same compounds, further validating the *in vivo* human permeability data set, as does the observation that passive diffusion is the dominant mechanism for the *in vivo* intestinal absorption of many drugs (Fagerholm et al. 1995; Winiwarter et al. 1998, 2003; Fagerholm et al. 1999). In this latter analysis the cut-off PSA value for low and high permeability was less than $100 \text{\AA}^2$ (high $P_{\text{eff}} > \approx 1.0 \times 10^{-4} \text{ cm s}^{-1}$ and $f_a > 90\%$). Some of these 22 investigated compounds were omitted from the final analysis in the detailed multivariate data analysis as they are carrier-mediated (amoxicillin, D-glucose, L-leucine, L-dopa, and α -methyl dopa) or paracellularly transported (urea and creatinine). The remaining 15 drugs were included in the multivariate analysis for passive membrane diffusion even if some of them (verapamil, losartan, furosemide and fluvastatin) have been considered to be substrates for efflux proteins, such as Pgp and multidrug-resistant protein (MRP), located in the enterocyte membrane. The intestinal absorption mechanism for these four model drugs is probably expected to be dominated by passive transcellular diffusion, which is supported by the fact that for these drugs, a linear relation exists between the fraction dose absorbed and the clinical dose range.

Regional differences in human *in vivo* permeability for five different compounds have been reported in a study that used an open single-pass triple-lumen tube and perfused an 80-cm-long segment of human jejunum and ileum (Riley et al. 1992a, b). The P_{eff} for

hydrochlorotiazide, atenolol, furosemide and cimetidine, all of which are classified as low-permeability drugs according to the BCS, decreased in the ileum in comparison with the jejunum (Lennernas 1998). This *in vivo* observation is in agreement with regional Ussing chamber studies in rats for low-permeability compounds (Ungell et al. 1998). Regional permeability coefficients of 19 drugs with different physicochemical properties have been reported using excised segments from three regions of rat intestine; jejunum, ileum, and colon. The *in vitro* permeability from rat small intestinal and colonic tissue mounted in an Ussing Chamber model clearly showed that the low-permeability drugs (BCS Classes III and IV) have an even lower permeability in the colon, whereas high-permeability drugs (BCS Classes I and II) show a slightly higher permeability in the colon when passive diffusion is the dominant transport mechanism (Ungell et al. 1998; Berggren et al. 2003; Van de Kerkhof et al. 2006). There was a significant decrease in the permeability of hydrophilic drugs and a significant increase in permeability of hydrophobic drugs aborally to the small intestine ($p < 0.0001$). Salicylic acid, which is highly permeable, was well absorbed throughout the rat small intestine. These animal data are considered as being very accurate and will predict human intestinal drug absorption, since several reports have demonstrated a high species correlation of intestinal permeability and absorption for drugs with passive diffusion and carrier-mediated transport mechanisms (Berggren et al. 2003, 2007; Cao 2007; Seithel et al. 2006; Van de Kerkhof et al. 2006). The small intestinal regional permeability pattern in humans has also been demonstrated for ranitidine (low human intestinal permeability drug) as well as paracetamol and grizeofulvin (two high intestinal permeability drugs), using a open single-pass intestinal perfusion technique (Gramatte 1994; Gramatte et al. 1994; Gramatte and Richter 1994; Lennernas 1998). *In vivo* permeability measurements of drugs in the colonic/rectal region in humans are difficult, which probably accounts for the limited quantity of published data. However, the perfusion of a defined and closed segment in the distal colon/rectum has been used in a few studies. It was observed that the permeability of antipyrine in the rectal region was high but D-glucose was not absorbed, which is in accordance with the fact that passive diffusion is the dominant drug absorption mechanism in this specific distal region of the human colon (Lennernas et al. 1995b, 2002a). Recently, it was reported that local administration of drugs and/or enzyme inhibitors in the human gut is possible by applying a small-bore, smooth tube that is introduced through the nose, retrieved from the pharynx. It is equipped with a firm radio-opaque capsule which is swallowed. Peristalsis moves the capsule to the desired location in the gut where it is anchored before drug administration via the tube. Drug uptake is followed by plasma sampling and one capsule type is used for solutions, another for solid formulations. With solutions, repeated administrations could be done with the capsule being anchored for 24 h or longer or, alternatively, at several locations along the gut (Petri et al. 2006a; Nyberg et al. 2007).

Assessing the effect of intestinal metabolism on P_{eff} is a methodological issue and has to include a study to establish which enzyme(s) is (are) involved and the site of metabolism in relation to the site of the measurements. Intracellular metabolism in the enterocyte, for instance by CYP3A4 and di- and tripeptidases, does not occur in the vicinity of the outer leaflet of the apical membrane, and is therefore not considered to affect the P_{eff} determined by the disappearance approach (single-pass perfusion) (Sandstrom et al. 1998a, b, 1999b; Sandstrom and Lennernas 1999; Tannergren et al. 2004). By contrast, drug metabolism in the lumen and/or at the brush border will directly interfere with the determination of the P_{eff} , since here, the drug is metabolized before it is absorbed (Kron Dahl et al. 1997, 2000). It has also been suggested that intracellular metabolism may indirectly affect intestinal permeability by providing a further sink boundary condition across the apical membrane. However, it has been reported that a specific inhibition and induction of human enterocyte

CYP3A4 by ketoconazole and St John's wort does not change the P_{eff} of *R/S*-verapamil, which suggests that the sink condition *in vivo* is provided by the highly perfused mesenteric blood vessels (Sandstrom et al. 1999b; Tannergren et al. 2004).

In accordance with its polar nature, furosemide ($\log D_{6.5} = -0.5$, $\text{PSA} = 124 \text{ \AA}^2$, $\text{HBD} = 4$) has a low jejunal P_{eff} of $0.05 \pm 0.014 \text{ cm s}^{-1}$ (Loc-I-Gut), and is classified as a low-permeability compound according to the BCS (Amidon et al. 1995; Winnivarter et al. 1998). Interestingly, after oral administration the absorption and bioavailability of furosemide are highly variable (35–75%). There are several hypotheses for this high variability such as active intestinal secretion, low passive diffusion, highly pH-dependent dissolution and permeability. A correlation between human jejunal P_{eff} and physicochemical descriptors suggests that drugs with octanol/buffer partitioning coefficients higher than zero and a $\text{PSA} < 100 \text{ \AA}^2$ will be highly permeable ($P_{\text{eff}} > \approx 1.0 \times 10^{-4} \text{ cm s}^{-1}$ and $f_a > 90\%$) across the human jejunum, and suggests that a strong influence of pH on physicochemical properties which will most certainly influence the passive intestinal permeability *in vivo*. At pHs 7.4, 6.5 and 5.5 the experimentally determined partitioning coefficients for furosemide at different pH were -0.9 , -0.5 and 0.4 , respectively (Winiwarter et al. 1998). In addition, uncharged furosemide (an acid with pK_a at 3.3/10.1) has an experimentally determined partition coefficient (i.e. $\log P$ value) of 2.53 ± 0.01 (Winiwarter et al. 1998). These strong pH-dependent properties might also explain why there is a higher fraction of the dose absorbed after oral administration than the determined human P_{eff} at pH 6.5. Finally, a recent *in vitro* study shows that active intestinal secretion process is important for the transport of furosemide across Caco-2 monolayers (Flanagan et al. 2002a, b). However, this is a good example where *in vitro* results have to be supported by *in vivo* data before any conclusions can be drawn regarding the mechanisms involved in the intestinal absorption of furosemide in humans.

Carrier protein expression

Regional intestinal differences in the functional activity of Pgp have only been reported in a few cases and are mainly based on animal tissues *in vitro*. For example, one of the few systemic kinetic analyses showed marked differences of *in vitro* efflux activity in tissue specimens originating from the different regions of the rat gastrointestinal tract (Makhey et al. 1998; Stephens et al. 2001, 2002a). The maximal *in vitro* transporter activity varied over a four- to fivefold range in the ileum > jejunum > colon (Stephens et al. 2001). It has been reported that MDR1 mRNA levels should be highest in the human colon (Thiebaut et al. 1987; Fricker et al. 1996; Thorn et al. 2005), but recent investigations report a higher expression of breast cancer resistance protein (BCRP/ABCG2), P-glycoprotein (Pgp/MDR1/ABCB1) and multidrug resistance protein 2 (MRP2/ABCC2) in the small intestine (highest expression in the jejunum) than in the colorectal region (Nakamura et al. 2002; Cao et al. 2006; Englund et al. 2006, 2007; Berggren et al. 2007; Hilgendorf et al. 2007). Nakamura et al. (2002) reported that MDR1, MRP1 and CYP3A mRNA levels were higher in human duodenal tissue than in normal colorectal tissue and colorectal adenocarcinoma tissue. For Pgp, the mRNA levels appear to be slightly lower in the region nearest the colon, but there was considerable variability between individuals and no statistically significant difference was observed between any of the intestinal regions. Hilgendorf et al. (2007) recently reported that of the 36 transporters investigated they found that 26 (69%) were expressed in jejunal mucosa and 21 (58%) in colon. In liver and kidney tissues 94 and 89% of the 36 transporters were expressed, respectively (Hilgendorf 2007).

This tissue difference is expected as the liver and kidney are extensively involved in transport of metabolism of most xenobiotics and endogenous compounds. The expression of BCRP/ABCG2, (Pgp/MDR1/ABCB1, and MRP2/ABCC2 in the gut mucosa of patients with ulcerative colitis showed that Pgp and BCRP were expressed at strongly reduced levels during active inflammation (Englund et al. 2007). Among the solute carriers the well characterized di- and tri-peptide carrier PepT1 were highly expressed in the human jejunum (Sun et al. 2002). The expression of PepT1 was very low in human colonic tissue that agrees with the low absorption of drugs such as amoxicillin from this intestinal region (Barr et al. 1994; Sun et al. 2002). All together, these studies demonstrate that the most important uptake (SLC transporters) and efflux carriers (ABC transporters) are largely expressed in the small intestine. The discrepancy in the results from different reports may arise from the study design, the patient population, the sampling techniques, the analytical methods, or any combination of these. Therefore, research efforts to examine the importance of these carriers on the overall drug absorption are most encourage to be performed in the proximal jejunum as this region has the highest expression of most carriers. This also justify the use of human *in vivo* permeability values determined in this region for prediction of fraction oral dose absorbed (Amidon et al. 1995; Fagerholm et al. 1996; Lennernas 1997, 1998; Sun et al. 2002; Cao et al. 2006).

Carrier-mediated intestinal absorption

Obviously, proximal small intestine is the tissue region where the absorption of nutrients and drugs are most rapid due to the surface area and the large number of carrier proteins, channels, and enzymes expressed (Sun et al. 2002; Lindell et al. 2003; Cao et al. 2006; Hilgendorf et al. 2007). Carrier-mediated intestinal absorption of drugs becomes the dominating mechanism for drugs with a reasonable high affinity for any intestinal transport protein, whereas the passive permeability contribution is small due to the polar nature of the compound. Targeting poorly absorbed drugs to these nutrient transport systems is a successful approach to increasing oral bioavailability (Lennernas et al. 1993; Han et al. 1998; Steffansen et al. 2005). The intestinal epithelium is polarized and many transport proteins are located in the apical membrane, and are maintained there because of the tight junctions, which prevents proteins from diffusing within the membrane. Tight junctions also prevent backflow of nutrients from the basal side of the enterocytes into the gut lumen. The two most important nutrient absorption proteins that can be utilized for drug transport in the absorptive direction are the oligopeptide carrier and the amino acid transport family. These proteins are only expressed in the small intestine to any significant degree and drugs, which are mainly absorbed through these carriers, will not be absorbed in the colon (Barr et al. 1994; Fagerholm et al. 1997; Lennernas 1998; Sun et al. 2002; Cao et al. 2006).

The oligopeptide carrier (hPepT1) is a symport carrier, which transports a substrate with a proton across the apical enterocyte membrane. Intestinal protein digestion generates a large variety and quantity of short chain peptides that are absorbed into intestinal epithelial cells by the hPepT1 transporter. Human hPepT1 operates as an electrogenic proton/peptide symporter with the ability to transport essentially every possible di- and tripeptide. Transport is enantioselective and involves a variable proton-to-substrate stoichiometry for uptake of neutral and mono- or polyvalently charged peptides (Daniel 2004; Daniel and Kottra 2004; Steffansen et al. 2005). Neither free amino acids nor peptides containing four or more amino acids are accepted as substrates (Friedman and Amidon 1991; Daniel 2004;

Daniel and Kottra 2004; Steffansen et al. 2005). The structural similarity of a variety of drugs with the basic structure of di- or tripeptides explains the transport of aminocephalosporins and aminopenicillins, selected angiotensin-converting inhibitors, and amino acid-conjugated nucleoside-based antiviral agents by hPepT1. The high transport capacity of hPepT1 allows rapid and efficient intestinal uptake of drugs but also of amino acid nitrogen even in states of impaired mucosal functions (Daniel 2004; Daniel and Kottra 2004). The access to protons of the oligopeptide carrier is a result of a sodium carrier, the Na^+/H^+ exchanger, which is located in the brush border membrane of the enterocytes (Daniel 2004). The jejunal contents of humans or animals, recovered after administration of model proteins such as albumin, showed that, the major fraction consisted of peptides typically with three to six amino acid residues, which corresponds to a concentration of 120–145 mM. The concentration of individual free amino acids in the digest varied between 0.6 and 16 mM, with a total concentration of all amino acids of 30–60 mM. Protein digestion and the absorption of the end products are generally not considered to be limited, and even quantities of 320–480 g of protein equivalent to 1.5–2.8 kg of lean meat eaten by human volunteers within 8 h did not yield evidence for a limitation in protein assimilation (Daniel 2004). In humans, the intestinal absorption of amoxicillin on average decreases from $72\% \pm 9\%$ to $45\% \pm 11\%$ if there is an increase in the oral dose of 500 to 3000 mg (Paintaud et al. 1992; Sjoval et al. 1992). This confirms that hPepT1 has a high transport capacity since amoxicillin is fairly good absorbed despite such large doses, but a moderate affinity to hPepT1. The *in vivo* jejunal P_{eff} for amoxicillin has been reported to be $0.4 \times 10^{-4} \text{ cm s}^{-1}$ at a perfusate concentration of 300 mg l^{-1} (0.82 mM), which corresponds to an oral dose of 1200 mg (Lennernas et al. 2002b). This jejunal P_{eff} value predicts that $f_a < 90\%$, which classifies amoxicillin as a low-permeability drug according to the biopharmaceutical classification system. However, the determined P_{eff} was higher than that predicted for amoxicillin from its physicochemical properties ($\log D_{6.5} = -1.7$, $\text{CLOGP} = 0.33$, $\text{PSA} = 154 \text{ \AA}^2$, $\text{MW} = 365 \text{ Da}$) (Winiwarter et al. 2003). This observation supports earlier pharmacokinetic reports, which suggests that amoxicillin has a higher intestinal absorption (about 50–75%) than expected for its low lipophilicity and amphoteric nature (Sinko et al. 1991). The large interindividual variability in the *in vivo* P_{eff} for amoxicillin might be due to polymorphism in the expression of hPepT1. Another possible explanation may be due to the nutritional status, since it has been reported that transcriptional activation of the PepT1 gene occurs by selective amino acids and dipeptides in the diet. It has also been reported that the integrated response to a certain stimulus may increase PepT1 activity by translocation from a preformed cytoplasmic pool (Lennernas et al. 2002b).

Enalaprilate, the diacid and active form of the drug, has a low P_{eff} and, consequently, a low f_a in humans (Table I and Figure 2). The prodrug approach of esterifying enalaprilate to enalapril increases the human *in vivo* P_{eff} as well as the f_a after oral administration (Figure 4) and is most likely due to the higher transport activity of hPepT1 with the ester prodrug enalapril than the diacid form of enalaprilate (Lennernas et al. 1994; Han et al. 1999; Sandstrom et al. 1999a; Sun et al. 2002). The human jejunal P_{eff} for amoxicillin, cephalexin, enalapril, and lisinopril have been reported (Takamatsu et al. 1997; Lennernas et al. 2002b; Sun et al. 2002). It has also been shown that in Caco-2 cells the *in vitro* permeability is low, which is in accordance with the poor expression of hPepT1 in that absorption model (Sun et al. 2002).

The *in vivo* P_{eff} in healthy volunteers of drugs transported by the amino acid transporter for large neutral amino acid (LNAA) has been determined for L-dopa and α -methyldopa. α -Methyldopa is classified as low P_{eff} drug, $0.1 \pm 0.1 \times 10^{-4} \text{ cm s}^{-1}$, at a perfusate

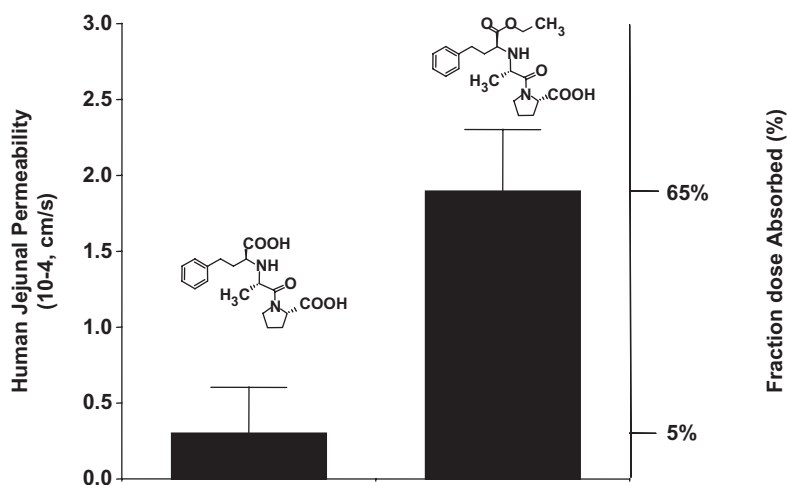


Figure 4. Effective *in vivo* jejunal permeability (P_{eff} mean $\pm$ standard deviation (SD)) of enalaprilat (left) and enalapril (right). The P_{eff} values predict the fraction dose absorbed for both drugs obtained in pharmacokinetic studies in humans. The higher jejunal P_{eff} of enalapril is most likely due to a significant higher transport through the peptide carrier (hPepT1). Data are from Winiwarter et al. (1998); Sun et al. (2002); and Chiu et al. (2003).

concentration of 6.0–6.5 mM (Lennernas et al. 1993; Nilsson et al. 1994; Lennernas 1998; Winiwarter et al. 1998). The corresponding *in vivo* P_{eff} for L-dopa is about 30 times higher, namely, $3.4 \pm 1.0 \times 10^{-4} \text{ cm s}^{-1}$, at a luminal concentration of 2.0–2.5 mM (Lennernas et al. 1993; Nilsson et al. 1994). The difference regarding *in vivo* P_{eff} between α -methyldopa and L-dopa is probably due to a lower affinity for and/or transport capacity for α -methyldopa of the LNAA transporter and suggests narrow substrate specificity for this transport route. The low *in vivo* P_{eff} of α -methyldopa indicates that passive diffusion is also low, which is in accordance with the physicochemical properties of the compound (MW = 211 Da, $\log D_{6.5} < -2$, PSA = 103 Å², HBD = 5) (Winiwarter et al. 1998). Two nutrient substrates for the above carrier family, L-leucine and L-phenylalanine, have high *in vivo* P_{eff} in humans even when determined at two widely different perfusate concentrations, which confirm that the LNAA amino transport family has a high *in vivo* transport capacity in the human jejunum (Lennernas et al. 1993; Takamatsu et al. 1997).

The rate and extent of intestinal absorption of cimetidine has been widely discussed previously, and a reasonable value of f_a for this drug has been estimated to be 75% (Grahnen et al. 1979; Grahnen 1984). It has been reported that cimetidine is a substrate for both Pgp and/or organic cation transporters (OCNT1 and OCNT2) (Collett et al. 1999; Takamatsu et al. 2001; Sun et al. 2002). The human jejunal *in vivo* P_{eff} is $0.22 \pm 0.13 \times 10^{-4} \text{ cm s}^{-1}$ $0.32 \pm 0.18 \times 10^{-4} \text{ cm s}^{-1}$ at luminal concentration of 250 and 667 mg l⁻¹, respectively (Figure 5). The lack of a difference between the two concentrations and together with the observation that the human permeability correlates well with that of the Caco-2 model suggests that passive diffusion is a dominating mechanism even for cimetidine. If organic cation transporters would have been the dominating mechanism, it could have been expected that the human intestinal *in vivo* permeability would have been significantly higher than the Caco-2 permeability since the expression of OCNT1 and OCNT2 is higher in the human small intestine than it is in the Caco-2 model (Takamatsu et al. 2001; Sun et al. 2002).

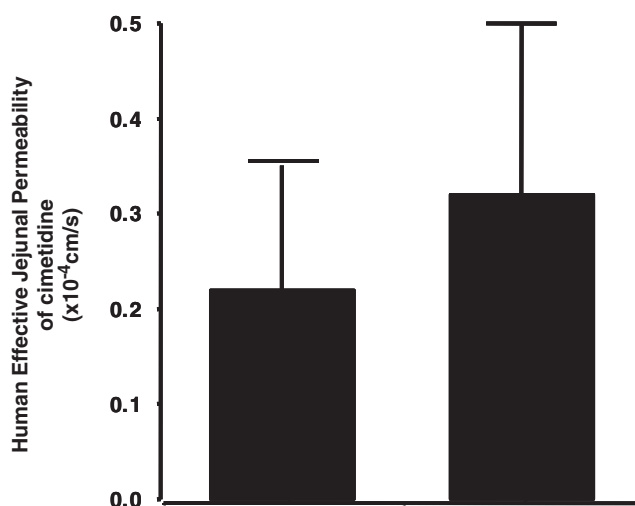


Figure 5. Effective permeability (P_{eff} , mean $\pm$ standard deviation (SD)) of cimetidine in human jejunum at two clinical relevant luminal concentrations 250 mg l^{-1} (left) and 667 mg l^{-1} (right), respectively. The rate and extent of intestinal absorption of cimetidine has been widely discussed previously, and a reasonable value of f_a for this drug has been estimated to be 75%. Data are from Winiwarter et al. (1998); Takamatsu et al. (2001); Sun et al. (2002); and Chiu et al. (2003).

Intestinal transport and metabolism

The most abundant isoform among human P450 enzymes is CYP3A4, which corresponds to 30% of the total P450 content in the liver and about 70% of the total P450 content in the intestine (Paine et al. 1997). The CYP3A4 isoform is one of the most important enzymes for oxidative drug metabolism and about 50–60% of clinically used drugs are metabolized by this particular isoenzyme (Smith and van de Waterbeemd 1999). Despite the higher CYP3A4 enzymatic capacity in the liver than in the small intestine, it has been shown that the human intestine contributes significantly to the first-pass extraction of drugs subjected to metabolism by CYP3A4, including verapamil, midazolam, budesonide and felodipine (Regardh et al. 1989; Wang et al. 1989; Gorski et al. 1998; Sandstrom et al. 1999b; Benet and Cummins 2001; Petri et al. 2006a).

The proposed apical recycling hypothesis is one attempt trying to explain this fairly high gut wall extraction in spite of the relatively low intestinal CYP3A4 activity in comparison with the liver. This hypothesis suggests that Pgp and CYP3A4 act synergistically to prolong the intracellular residence time and thereby repeatedly expose the drug to the CYP3A4 enzyme. In addition, the process may provide an active transport of formed metabolites towards the intestinal lumen which may result in product inhibition of the enzyme. Further evidence of the metabolism–efflux interplay is the proposed co-localization of Pgp and CYP3A4 in the enterocyte and the overlapping substrate specificity (Benet and Cummins 2001). However, strong *in vivo* evidence to support this elegant hypothesis is still lacking, and its validity *in vivo* is not evident as CYP3A4 substrates, such as midazolam and felodipine, undergo extensive gut wall metabolism even though they are not subjected to any intestinal efflux (Regardh et al. 1989; Wang et al. 1989; Shen et al. 1997; Gorski et al. 1998).

Recently, several clinical studies have also claimed that an inhibition of intestinal efflux (especially of Pgp) is the major cause behind an increased bioavailability when certain drugs

were co-administered (Fricker et al. 1996; Floren et al. 1997; Lown et al. 1997; Christians et al. 2002; Kruijtzter et al. 2002a, b; Glavinas et al. 2004; Kunta and Sinko 2004). However, in many of these latter reports the most likely main mechanism was still the inhibition of CYP3A4 and the role of the cellular efflux *in vivo* at the intestinal level remains unclear (Tamura et al. 2002, 2003; Lin 2003; Lin and Yamazaki 2003a, b; Lindahl et al. 2004). One such example is paclitaxel that has been reported to have a low bioavailability following oral administration mainly due to intensive intestinal efflux (Bardelmeijer et al. 2002b; Kruijtzter et al. 2002a, b; Stephens et al. 2002a, b). Several inhibitors originally claimed to inhibit Pgp most likely increased the bioavailability by first-pass metabolism inhibition (Schellens et al. 2000; Peltier et al. 2006). However, the main reason for paclitaxel low bioavailability seems to be a combination of its poor aqueous solubility, low *in vivo* dissolution rate and extensive metabolism (Bardelmeijer et al. 2002a; Peltier et al. 2006). Increased solubility and dissolution rate will also provide higher luminal concentration that might more efficiently saturate intestinal efflux and/or metabolism for paclitaxel (Sandstrom et al. 1999a; Peltier et al. 2006). It has also reported that paclitaxel has a relatively high *in vitro* permeability (a-b) in Caco-2 cells ($4.4 \pm 0.4 \times 10^{-6} \text{ cm s}^{-1}$), which suggests that the drug will be effectively absorbed in the intestinal tract (Walle and Walle 1998). The reason for this over interpretation of the role of enterocyte efflux activity on intestinal drug absorption of paclitaxel and other extensively metabolized efflux substrate may be due to the misinterpretation of the role of first-pass metabolism and the specificity of the inhibitor used. It is also important to recognize that paclitaxel is transported by Pgp but it has no quantitative importance, which is in accordance with the BDDCS classification of drugs undergoing almost complete metabolism to be classified as high-permeability drugs (Wu and Benet 2005).

A direct *in vivo* assessment was undertaken with the single-pass perfusion approach in the human jejunum by using the Loc-I-Gut technique with *R/S*-verapamil ($\log D_{6.5} = 2.7$, octanol/aqueous buffer pH 6.5; pH 7.4; MW = 455 Da) as the model compound for CYP3A4 and Pgp-mediated local intestinal kinetics. The determined P_{eff} for both enantiomers of verapamil at perfusate concentrations of 4.0, 40, 120 and 400 mg l^{-1} were 2.5×10^{-4} , 4.7×10^{-4} , 5.5×10^{-4} , and $6.7 \times 10^{-4} \text{ cm s}^{-1}$, respectively (Sandstrom et al. 1998b, 1999b; Tannergren et al. 2004). A luminal concentration of 400 mg l^{-1} is expected to be achieved in the upper part of the small intestine after oral administration of a 100 mg dose of verapamil in an immediate release dosage form. The three other perfusate concentrations represent fractions of the dose when 30, 10, and 1%, respectively, remain to be absorbed. The increased *in vivo* jejunal P_{eff} of *R/S*-verapamil along with its increased luminal perfusate concentration is in accordance with a saturable efflux mechanism mediated by Pgp. There was no difference in P_{eff} between the *R*- and *S*-forms of verapamil at any luminal concentration, which suggests that the efflux transport cannot discriminate between the *R*- and *S*-forms of verapamil. However, the measured *in vivo* jejunal P_{eff} was sufficiently high at all four perfusate concentrations ($>2.0 \times 10^{-4} \text{ cm s}^{-1}$) to predict complete absorption of the given dose (Table I and Figure 2). Taken collectively, with the fact that P_{eff} and f_a are well predicted from the physicochemical properties of *R/S*-verapamil, this suggests that passive diffusion is the dominating transport mechanism for this drug in the human intestine. These were among the first investigations to challenge the role of Pgp as a general absorption barrier along the intestine for drugs which are substrate for any efflux carrier (Sandstrom et al. 1998a, b, 1999b).

Ketoconazole, a well-known potent inhibitor of CYP3A4 metabolism and a less potent Pgp modulator, acutely inhibited CYP3A4 metabolism but did not affect the P_{eff} of *R/S*-verapamil when they were co-perfused through the human jejunal segment,

ketoconazole at 40 mg l^{-1} and *R/S*-verapamil at 120 mg l^{-1} (Table I and Figure 2) (Sandstrom et al. 1999a, b). This confirms that, *in vivo*, ketoconazole is less potent as an inhibitor of Pgp in humans than is CYP4A4, even if verapamil is transported by passive diffusion. To a large extent, an increased P_{eff} would have been expected for a Pgp inhibition. Other examples are the jejunal P_{eff} for cyclosporine and tacrolimus, well-known CYP3A4 and Pgp substrates, which have been reported to be high and to predict a complete intestinal absorption, even if the drug is subjected to significant efflux *in vitro* (Sakaeda et al. 2001; Tamura et al. 2002, 2003; Chiu et al. 2003; Cao et al. 2006). Taken collectively, these directly determined *in vivo* human and rat P_{eff} values suggest that drugs with a high passive intestinal permeability contribution to the overall absorption rate will be completely absorbed from the gut even if they are substrates. This may be due to a high contribution of passive diffusion and that these intestinally located efflux proteins may be easily saturated due to high drug concentrations within the intestinal membrane.

Model drugs and direct *in vivo* methods are needed in order to perform accurate investigations of the clinical significance and the pharmacogenetics of transporters and their influence on pharmacokinetics. Such model compounds should not be subjected to any metabolism and accordingly have a hydrophilic nature with low passive diffusion, which would make the assessment of the role of transporter possible. For instance, talinolol, digoxin, and fexofenadine have been suggested as model compounds to assess the phenotype for Pgp significance (Figures 6 and 7). The plasma exposure of oral digoxin was significantly lower during rifampin treatment, a finding which was attributed to an increased expression of intestinal Pgp (Figure 6). Recently it was reported that quinidine increased the rate of jejunal absorption and systemic availability of digoxin when applying a multi-lumen perfusion catheter with two 20-cm adjacently isolated jejunal segments. This effect is most likely due to an inhibition of the apical efflux of digoxin (Igel et al. 2007). Even if digoxin is a good clinical example it might be a special case since it is given at low clinical dose (often less than 1.0 mg). A direct comparison of the importance of Pgp and CYP3A4 in the gut is possible when comparing the effect of pretreatment with St John's wort (12 days of 900 mg or 14 days of 300 mg three times per day) on the pharmacokinetics of talinolol and verapamil, respectively. Talinolol is a non-metabolized Pgp substrate, while verapamil is completely metabolized, mainly by CYP3A4 (Sandstrom et al. 1999a, b; Schwarz et al. 2007). St John's

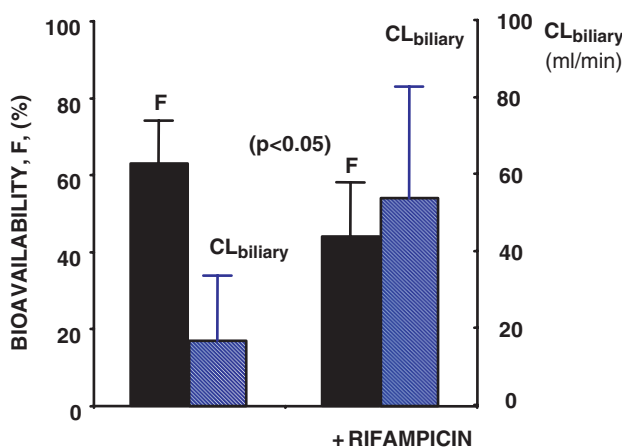


Figure 6. Effect of pre-treatment of rifampicin (600 mg o.d.) on the bioavailability and biliary secretion of digoxin in human subjects. Data are from Greiner et al. (1999).

wort markedly increased both CYP3A4-mediated metabolism and Pgp transport and plasma exposure decreased only by about 25–30% for talinol and by 80% for verapamil (Tannergren et al. 2004; Schwarz et al. 2007). This also suggests that CYP3A4 has a more prominent role than intestinal Pgp in the reduction of bioavailability of orally administered drugs that are substrate for these proteins.

In vitro data based on Caco-2 cells suggest that the *in vitro* permeability of fexofenadine, a substrate for Pgp and organic anion transport protein (OATP), was increased in the absorptive direction by approximately two- to threefold (from approximately 0.3×10^{-6} to $1.5 \times 10^{-6} \text{ cm s}^{-1}$) in the presence of various Pgp-inhibitors (such as verapamil, ketoconazole and GF 120918) (Figure 8) (Cvetkovic et al. 1999; Petri et al. 2004; Glaeser et al. 2007). Hence, an expected acute effect of ketoconazole and verapamil when added to the jejunal perfusion would have been an increase in jejunal P_{eff} for fexofenadine, but there was no acute effect on the jejunal P_{eff} of fexofenadine. The jejunal P_{eff} was low ($0.1\text{--}0.2 \times 10^{-4} \text{ cm s}^{-1}$) and variable, which, according to the BCS, classifies the drug as a low-permeability compound (Figures 2 and 8). It has also been reported that the plasma exposure of fexofenadine is linear over a wide dose range of 40–800 mg (Figure 9) (Russell et al. 1998). This is in agreement with the linear permeability in the absorptive direction for fexofenadine seen in the Caco-2 model (Figure 10) (Petri et al. 2004). The intestinal absorption process of fexofenadine is complex and the role of Pgp is not clear. Plasma data from drug–drug and food–drug interactions studies suggest that, however, there are specific uptake and efflux transporters in the liver that are involved in the pharmacokinetics of fexofenadine (Tannergren et al. 2003b; Glaeser et al. 2007). It has even been suggested that uptake transporters in the intestine should play a role, but this issue has not been investigated *in vivo* in human with a perfusion technique (Kikuchi et al. 2006; Glaeser et al. 2007).

Results obtained solely from cell culture studies might have led to an over interpretation of the importance of efflux carriers on the intestinal absorption process. Many drugs that were initially believed to undergo significant efflux *in vitro* were later shown to be completely

Fexofenadine

A suitable *in vivo* probe for transporter?

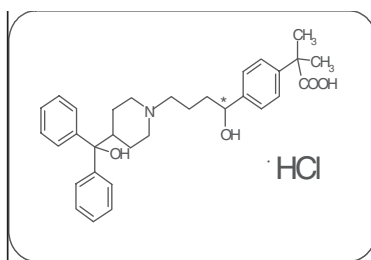
Low passive membrane permeability both *in vitro* and *in vivo*

In vitro transport by ABCB1 (Pgp), OATP2B1, 1A2 and 1B3

Low degree of *in vivo* metabolism

In vivo drug–drug and food–drug interactions – probably transporter mediated

Linear relation between oral dose and plasma exposure in the dose range 10–800 mg



Selective nonsedating histamine H1 receptor antagonist.
MW 538 g/mol (salt)
Zwitterion with pKa at 4.2 and 9.5
Solubility 3.6 mg/ml
Polar surface area 124 Å²

Figure 7. Proposed role of fexofenadine as a model drug for *in vivo* assessment of P-glycoprotein function.

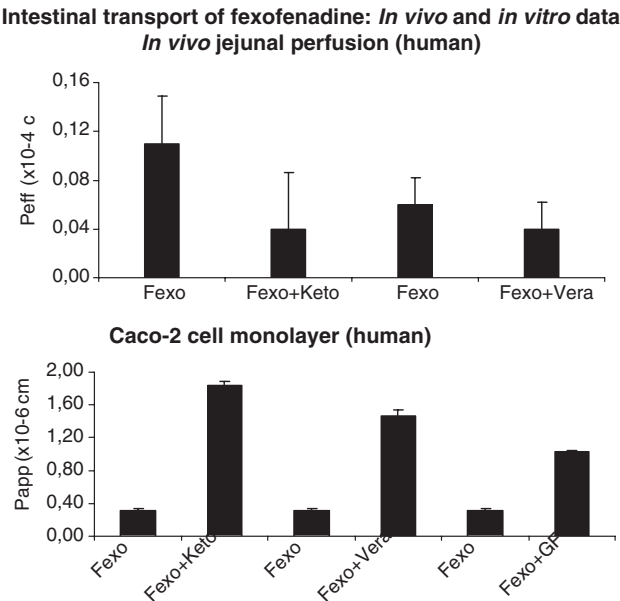


Figure 8. Effect of various inhibitors on the transport of fexofenadine in human jejunum (*in vivo*) and Caco-2 cell monolayers (*in vitro*). There is an effect in the Caco-2 model but not on the human *in vivo* permeability. Data are from Tannergren et al. (2003a, b); and Petri et al. (2004).

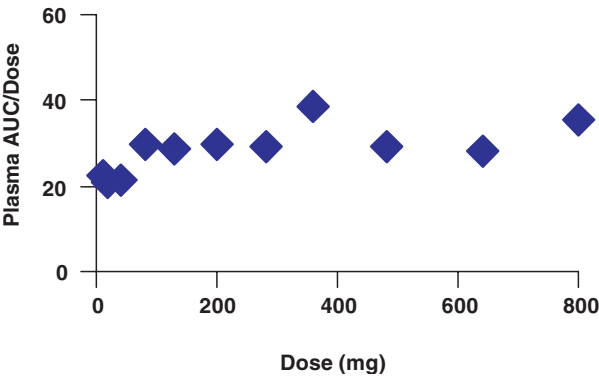


Figure 9. Linear relationship exists between the bioavailability and oral doses of fexofenadine. The oral dose range was between 40 and 800 mg. Data are from Russell et al. (1998).

absorbed *in vivo* (Cao et al. 2006). However, one of the important functions of efflux proteins in the intestine and the liver that belongs to the ABC transport family are to secrete hydrophilic conjugates formed intracellularly (Petri et al. 2003). They also affect the rate of distribution of drugs into the central nervous system.

Intestinal secretion as the elimination route

Direct intestinal secretion of drugs has been proposed as an elimination route and as a contributor to the total clearance of drugs (Gramatte et al. 1999). This is because many

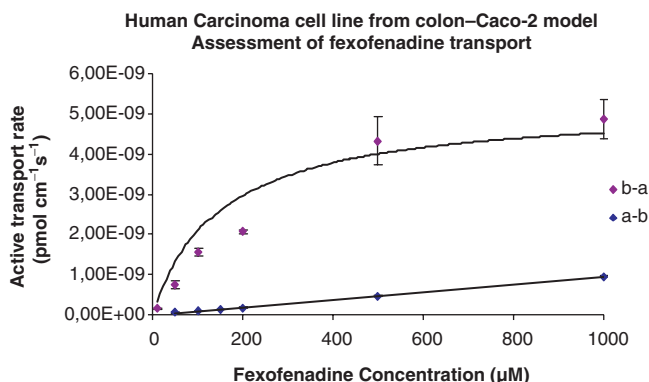


Figure 10. Assessment of fexofenadine transport in the Caco-2 model. Transport is linear in the absorptive direction (a-b), but saturable in the secretory direction (b-a). The apparent $V_{\max}$ and K_m were $5.21 \text{ pmol cm}^{-2} \text{ s}^{-1}$ and $150 \text{ } \mu\text{M}$, respectively, in the secretory direction (b-a). Data are from Petri et al. (2004).

efflux transporters located at the apical membrane of enterocytes might contribute to the secretion of drugs. In order to be quantitatively important, there has to be an extensive transport of drugs into the lumen and limited reabsorption since there a rather restricted blood flow to the mucosal cells in the small intestine exists at only 181 h^{-1} (300 ml min^{-1}) in humans (a 70-kg person). Hence, drugs using this elimination route from the blood to the lumen will most likely be hydrophilic in nature and therefore a low passive diffusion across both the apical and basolateral membranes of the enterocytes is expected. Consequently, in order for this elimination route to be quantitatively important, it is considered that carrier mechanism(s) also have to present at the basolateral and not only at the apical membrane.

In six healthy volunteers the intestinal steady-state perfusion technique (triple lumen tubing system) has been used to measure the appearance of talinolol within the small intestine while the drug was infused intravenously (Gramatte and Oertel 1999). Talinolol was transported into the intestinal lumen, generating a five times higher concentration than in blood. Co-perfusion of an *R*-verapamil-containing perfusion fluid resulted in lower secretion rates of talinolol, corresponding to 29–56% of the values obtained in the absence of verapamil supplied intraluminally. Intravenously administered talinolol is actively secreted into the human small intestine. This secretion is reduced by the intraluminal supply of the Pgp modulating drug *R*-verapamil.

The direct intestinal secretion of digoxin was reported from a study where an intestinal perfusion catheter in healthy volunteers where two 20-cm jejunal segments were used (Greiner et al. 1999). Within 3 h after intravenous administration of digoxin (1 mg), perfusate samples were collected. It was found that $0.45\% \pm 0.24\%$ and $0.83\% \pm 0.60\%$ of the digoxin dose were eliminated into a jejunal segment and into bile, respectively. Perfusion of the isolated segment with quinidine reduced intestinal digoxin elimination ($0.23\% \pm 0.08\%$, $p = 0.031$). In the presence of rifampin, intestinal digoxin elimination was 0.80 ± 0.59 ($p = 0.383$) (Greiner et al. 1999).

The local pharmacokinetics of fexofenadine and rosuvastatin have been investigated in the intestine and liver by the use of the pig as model for drug transport in the enterohepatobiliary system (Petri et al. 2006b). The pig has been used extensively in biomedical research and is commonly used for physiological studies of the digestive system. Pigs have received increased attention as potential models for investigating the impact of intestinal absorption

and the first-pass effects on drugs (Petri et al. 2006b). The expression and function of membrane transport proteins and enzymes have been explored less thoroughly in the pig than in the rat, but it has been reported that orthologues of ABCB1 and OATPs are present in the intestine and liver of pigs. These studies have investigated and compared the biliary clearance with the intestinal clearance. The bile area under the curve (AUC)₀₋₆ was 46–69 times higher than the portal plasma AUC₀₋₆ for fexofenadine in pig. The disposition of fexofenadine was investigated in an intravenous study in combination with sampling of both bile and intestinal secretion. The intestinal secretion clearance (scaled up to the whole intestine) was $5.0 \pm 4.0 \text{ ml min}^{-1}$, biliary clearance was $62.1 \pm 45.3 \text{ ml min}^{-1}$, and total clearance for fexofenadine was $217 \pm 91 \text{ ml min}^{-1}$. This suggests that intestinal clearance plays a minor role in the total clearance of fexofenadine (Petri et al. 2006b).

The biliary secretion of rosuvastatin in healthy volunteers using an intestinal perfusion method after administration of 10 mg rosuvastatin dispersion in the intestine has been reported to be extensive (Bergman et al. 2006). At all time points the biliary concentration exceeded the total plasma concentration, and the average bile-to-plasma ratio was about 5000. In pig the bile AUC₀₋₆ was 1000–2000 times higher than the portal plasma AUC₀₋₆ for rosuvastatin. The disposition of rosuvastatin was investigated by applying a combination of sampling of both bile and intestinal secretion. The intestinal secretion clearance (scaled up to the whole intestine) was $1.5 \pm 0.7 \text{ ml min}^{-1}$ and biliary clearance was $245 \pm 195 \text{ ml min}^{-1}$. This suggests that intestinal clearance plays a minor role in the total clearance of rosuvastatin. One of the main explanations about why the fexofenadine (a Pgp substrate) and rosuvastatin (a BCRP substrate) have a low intestinal secretion might be the absence of an efficient transport from blood across the basolateral membrane into the enterocyte.

Conclusion

This review emphasized the need for more *in vivo* studies in order to clarify the dynamic interplay between mechanisms of drug transport and metabolism in the human intestine. There is also a need for further development of *in vivo* techniques for direct measurements of these processes from various regions along the human gastrointestinal tract and to relate this to various physiological/pathophysiological conditions. This would certainly increase knowledge about the important mechanisms involved as well as providing high-quality *in vivo* data for the development and validation of rapid and more reliable *in vitro* intestinal models as well as forming the basis for successful physiologically based modelling. The available human *in vivo* permeability data also clearly show that the intestinal epithelium has a large resistance towards large and hydrophilic compounds that are not a substrate for any uptake transporter and that the paracellular route has a low contribution for compounds larger than MW = 300. It was also the aim to offer a perspective of the role of intestinal efflux protein in rate and extent of intestinal absorption of drugs. It seems that the intestinal efflux proteins has a less dominant role to affect the extent of intestinal absorption and that intestinal efflux data based on *in vitro* models might be over-interpreted. Several drugs that are correctly assigned as intestinal efflux substrates *in vitro* instead have a low systemic bioavailability due to poor solubility and/or extensive first-pass metabolism. Several sufficiently lipophilic parent drug compounds have historically been reported to be Pgp substrates but since many of them are subjected to extensive metabolism, this also classifies them as high-permeability drugs according to the recently proposed BDDCS

(Wu and Benet 2005). To be a Pgp substrate so that its extent of absorption has been affected by intestinal efflux, the drug has to be given in low dose and have a low or modest passive diffusion.

Finally, for a better understanding of the *in vivo* relevance of membrane transporters on the pharmacokinetics and pharmacodynamics of drugs and metabolites, one needs to use other approaches, such as *in vivo* perfusion techniques and molecular imaging techniques, besides plasma concentration measurement. Microdosing is another clinical approach that has gained increased focus recently (Bergstrom et al. 2003). Since many new drug entities are more or less efflux substrates but also fairly lipophilic, the above discussion suggests that first-pass metabolism rather than the intestinal efflux problem should affect the validity of the microdosing concept.

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